COMBINATIONAL LOGIC DESIGN

Gate Perspective (Approach)

Modular Perspective (Approach)





The Building Blocks

1) Data Manipulation

2) - Code conversion

3) - Data selection from various sources

4) - Data Routing from source to various Destinations

5) Data Error detection

6) - Data busing from one part of the digital system to another

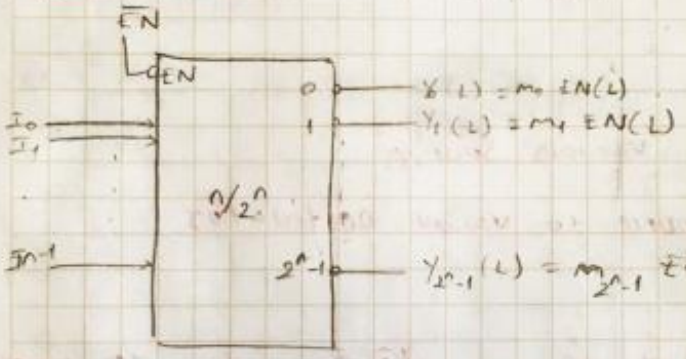
Arithmetic Circuit Type — Multipurpose Circuit

ADDER	DECODER - ENCODER
SUBTRACTOR	CODE CONVERTER
MULTIPLIER	MULTIPLEXER - DEMULTIPLEXER
DIVIDER	SHIFTER
COMPARATOR	ALU (Arithmetic and Logic unit)
PARITY GENERATOR	ROM (Read only memory)
ERROR DETECTOR	PLA (Programmable Logic Arrays)
	" Array Logic

DESIGN PROCEDURE

1- Understand the Device
2- State the Algorithm
3- Construct a truth Table
4- Obtain output Function
5- Construct the Logic Diagram
6- Check (Test) the Results

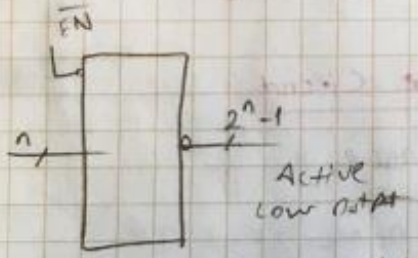
DECODER AND ENCODER

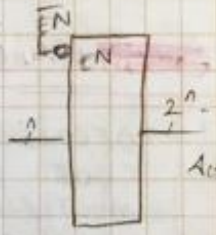


$m_i(L) = M_i(H)$

$M_i(L) = m_i(H)$

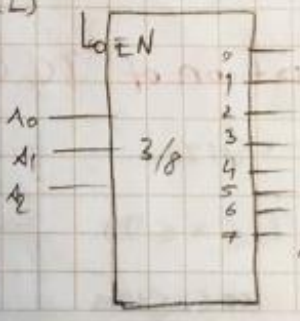
$y_i = m_i \overline{EN}(L)$





$\overline{EN}$	A_2	A_1	A_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	X	X	X	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	1	0	0
1	0	1	1	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0

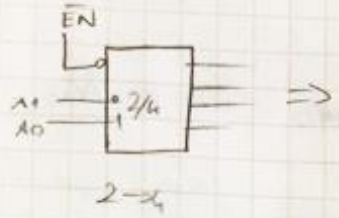
$y_i = m_i \overline{EN}(L)$



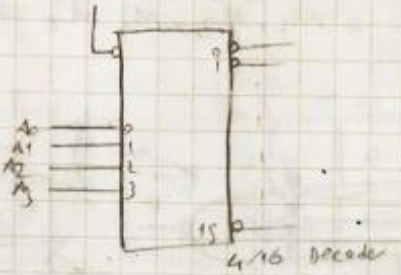
$\overline{A}_2 \overline{A}_1 \overline{A}_0 \overline{EN}(L)$

$A_2 A_1 A_0 \overline{EN}(L)$

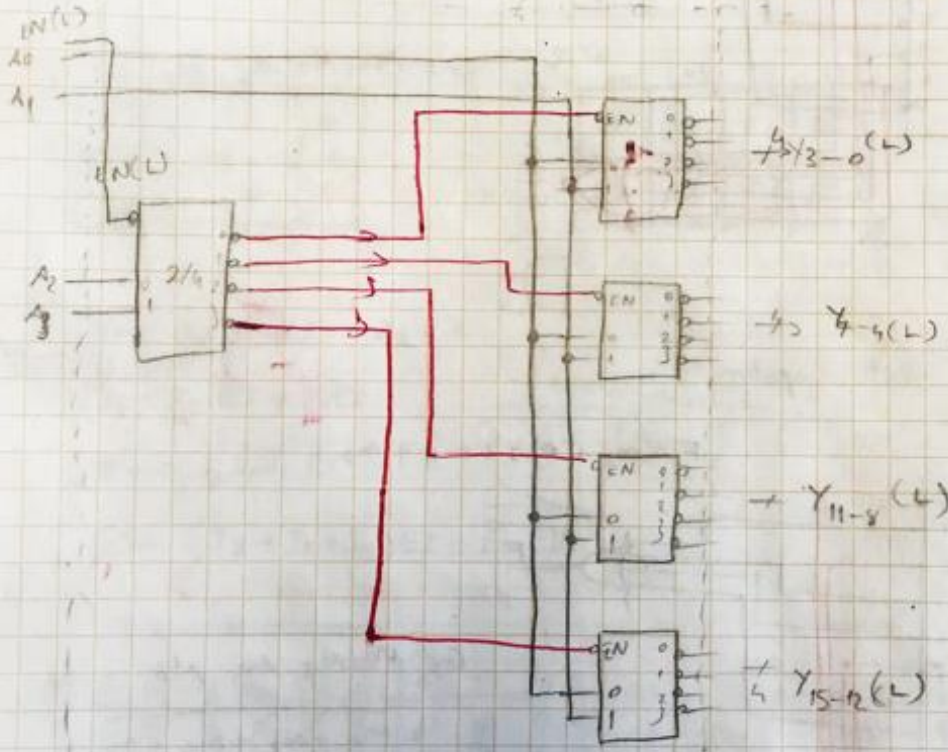
EXPANSION FORMAT



4→16 Decoder

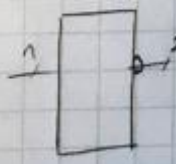


2/4 den 4/16 farke
elde etmek için



Combinational Logic Design with Decoder

Decoder





Logic Function

Canonical SOP form

Canonical POS form

Combinational Logic Design with Decoder

Decoder

n → 2^n inverting output

n → 2^n Non inverting

Logic function

Cannonical SOP form

Cannonical POS form

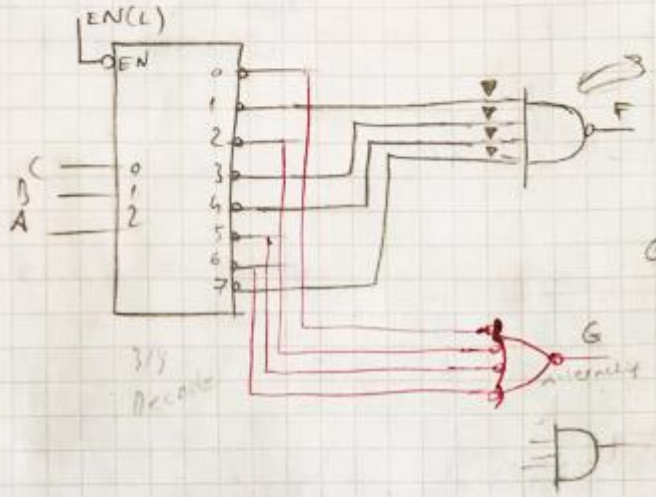
OSOW

$F(A,B,C) = \sum m(1,3,4,7)$

$G(A,B,C) = \prod M(0,2,5,6)$

$\overline{F} = M_1$

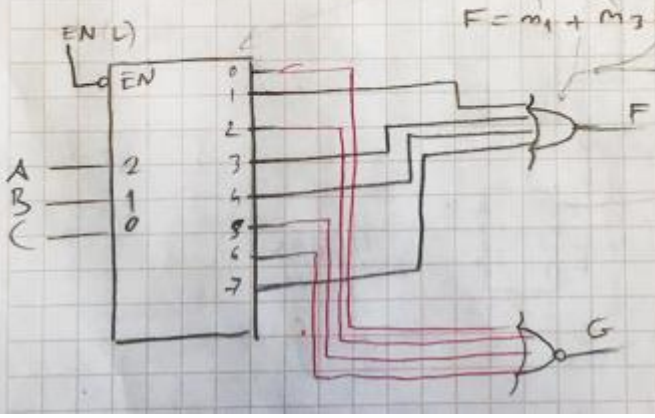
$F = m_1 + m_3 + m_4 + m_7 = \overline{M_1 \cdot M_3 \cdot M_4 \cdot M_7}$ NAND



$G = M_0 \cdot M_2 \cdot M_5 \cdot M_6$

Evirmeyen şekilli yapalım

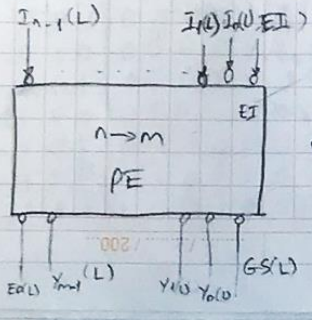
$F = m_1 + m_3 + m_4 + m_7$



$G = M_0 M_2 M_5 M_6$

$= \overline{m_0 + m_2 + m_5 + m_6}$

PRIORITY ENCODER (PE)



(Priority Encoder)

Enable Input

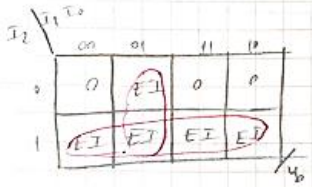
$n \leq 2^m$

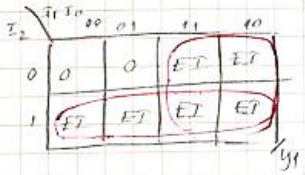
Design of a Priority Encoder.

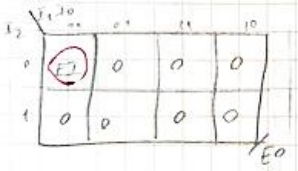
→ Group select

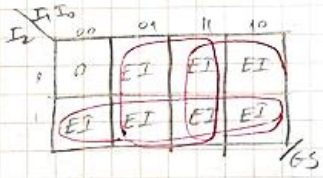
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Priority scheduel	$\overline{EI}$	I_2	I_1	I_0	GS	Y_1	Y_0	E_0
	0	X	X	X	0	0	0	0
I_2 (highest) Encoded to 11	1	1	X	X	1	1	1	0
I_1 " to 10	1	0	1	X	1	1	0	0
I_0 " to 01	1	0	0	1	1	0	1	0
Inactive else Assigned to 00	1	0	0	0	0	0	0	1









$Y_0 = \overline{I_1}\, I_0\, EI + I_2\, EI$

$Y_1 = I_2\, EI + I_1 EI$

$E_0 = \overline{I_2}\,\overline{I_1}\,\overline{I_0}\, EI$

$GS = (I_2 + I_1 + I_0) EI = \overline{E_0}\, EI$

$\overline{EI}$	I_5	I_4	I_3	I_2	I_1	I_0	GS	Y_1	Y_0	E_0
0	X	X	X	X	X	X	0	0	0	0
1	1	X	X	X	X	X	1	1	1	0
1	0	1	X	X	X	X	1	1	0	0
1	0	0	1	X	X	X	1	0	1	0
1	0	0	0	1	X	X	0	1	1	0
1	0	0	0	0	1	X	0	1	0	0
1	0	0	0	0	0	1	0	0	1	0
1	0	0	0	0	0	0	0	0	0	1

